COMSATS University Islamabad
Department of Management Sciences · BS (Business Data Analytics)

FINAL YEAR PROJECT

Configuration Control Dashboard for the Google Pixel 7

Automated configuration management with live drift detection on real public data

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Project Goals

1. Implement the **four core functions of ANSI/EIA-649** configuration management in one dashboard
2. Detect **configuration drift automatically** for every item and every variant
3. Trace **every change to an approved** Engineering Change Request (ECR)
4. Answer an **auditor in seconds**, not days
5. Prove it on the **real public history** of the Google Pixel 7

194	73	8,212	12	7	35	11	6	8	104
configuration items	real Pixel 7 builds	sourced versions	variants	data sources (6 real)	historical ECRs	baselines 2022–2026	drift rules	dashboard pages	tests passing

1 The Problem

A phone has **~200 tracked items** (chips, firmware, software, parts, certificates), sold in many variants, and something changes every month. Without control:

- Nobody knows exactly what is in each variant **right now**
- Changes slip in **without approval**
- Regions get **left behind** on old software
- Proving the configuration to an auditor **takes days**

2 The Solution

For every item and variant, the CCD compares **three** things:

APPROVED	ACTUAL	AUTHORIZED
Active baseline what it should be	GitHub + PLM what is deployed	Jira ECRs what was approved

↓ compare ↓

✓ Match, or covered by an approved ECR
Compliant: green, traceable to the ECR

● Differs with no approved ECR
Critical “unauthorized change” alarm

3 Change Control

Open → Impact Review → CCB Review → Approved → Promote baseline → Implemented
↳ Rejected: alarm stays

Promote **refuses anything unapproved**. Each approved baseline is sealed with a **SHA-256 fingerprint**, so tampering is detected.

4 Architecture

Factory images REAL
73 builds, bootloader + radio

AOSP source REAL
7,800+ commits, tags

Bulletins REAL
952 CVEs, patch levels

FCC, iFixit REAL
hardware facts

GitHub config REAL
ACTUAL software state

Jira ECRs REAL
AUTHORIZATION

Factory PLM (FastAPI) MOCK
ACTUAL hardware: parts, revisions, BOMs

↓

Sync idempotent, logged → **Database** 194 CIs, 12 variants → **Engine** baselines, rules, reports → **Dashboard** 8 pages, audit PDF

5 Features → EIA-649

Identification	CI register, variants, baselines **CI Register · Baselines**
Change control	ECR workflow, approved-only baselines **Change Requests**
Status accounting	KPIs, variant matrix, comparisons **Overview · Variant Matrix**
Verification & audit	6 rules, compatibility, sealed baselines **Drift · Compatibility · Audit**

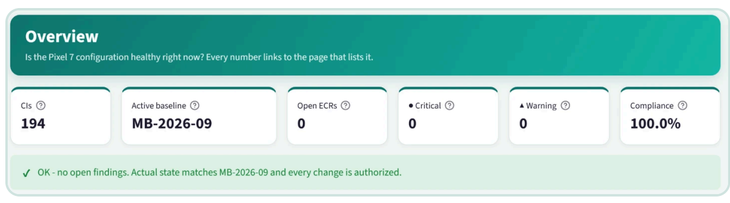
6 Six Drift Rules

- **Unauthorized change** — no approved ECR
- **Unimplemented change** — approved, not done in 7 days
- **Compatibility violation** — parts that do not fit
- **Variant inconsistency** — a model left behind
- **Baseline integrity** — sealed record tampered
- **Stale security patch** — older than 60 days

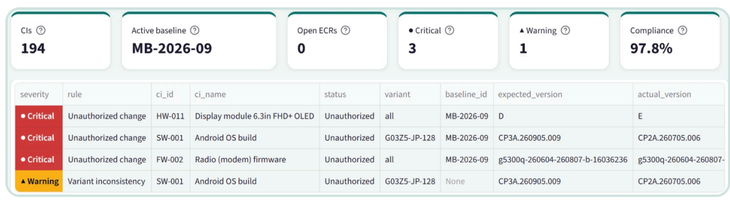
Critical · Warning

7 Results: Live Demo

BEFORE Baseline MB-2026-09: **0 findings, 100% compliance**



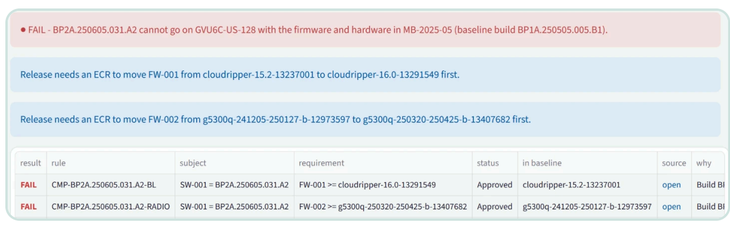
AFTER Unapproved radio firmware, Japan model left behind, new display revision: **3 Critical** **1 Warning**, each pinpointed to item and variant



MATRIX Variant matrix: **red = differs** from the approved baseline, per variant



BLOCKED First Android 16 build **blocked** on the last Android 15 baseline, using Google’s real bootloader and radio requirements



- Approving it in Jira **clears the alarm automatically**
- History rebuilt 2022–2026 via **35 ECRs**
- Live rehearsal: **3/3 runs passed**

8 Project Plan

1 Research	2 Data & Sync	3 Engine	4 Dashboard	5 Testing
EIA-649, public sources	schema, 4 connectors	baselines, 6 rules, compatibility	8 pages, audit PDF, auto-sync	104 tests, live rehearsal 3/3
✓ Done	✓ Done	✓ Done	✓ Done	✓ Done

All five phases complete: system **ready for live demonstration** at the final evaluation.

9 Deliverables

- ✓ CCD web dashboard (8 pages)
- ✓ GitHub, Jira, history and PLM sync
- ✓ Drift engine + compatibility check
- ✓ Audit report (CSV and PDF)
- ✓ Test suite and FYP report

10 Conclusion

The CCD answers in **seconds** what takes **days**: what is in every variant, was every change approved, and where did reality drift?

Python 3.12 · SQLAlchemy · FastAPI · Streamlit · Plotly · GitHub API · Jira API · ReportLab · pytest

Future work: real PLM connector, in-app approvals, multi-device support, webhooks.

Public-data case study. Software and firmware versions, build history, security bulletins and hardware facts are real public Google/AOSP/FCC/iFixit data. Part numbers, revisions, BOMs and approvals are mocked, as no phone maker publishes them. Not Google’s internal data.